MICHAEL IWANEK

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WORK EXPERIENCE

Full Stack Software Engineer **Jun 2023 - Present**
PNNL - Pacific Northwest National Laboratory Hybrid - Richland, WA

- Develop scalable backend services in .NET (C#) and Python for a distributed computer vision and machine learning platform supporting million-image datasets and simulation workflows. Architect asynchronous APIs and queue-backed microservices for parallel processing, long-running tasks, and image analysis. Integrate EF Core, GraphQL, and OpenSearch for efficient data retrieval and complex filtering.
- Architect PostgreSQL schemas for 100M+ records using partitioning and indexing for scalable performance. Develop custom release scripts for schema evolution, data migration, and index/constraint tuning. Offloaded filtering queries to OpenSearch, reducing latency from roughly 20 minutes to 200 ms and achieving subsecond filtering in billion-record scalability tests.
- Design asynchronous Kafka, SQS, and RabbitMQ messaging pipelines for decoupled, fault-tolerant communication across distributed systems. Implement consumer groups, retry logic, and dead-letter queues for parallel processing, dynamic scaling, and resilient failure recovery.
- Engineer and maintain GitLab CI/CD infrastructure, deploying and configuring self-hosted GitLab Runners. Develop reusable pipeline components to automate builds, testing, releases, and AWS deployments with CDK and CloudFormation. Containerize services with Docker and orchestrate large-scale end-to-end tests through child pipelines, CI/CD variables, and shell scripts.
- Develop multithreaded Python microservices and Apache Airflow DAGs for data ingestion, database operations, and infrastructure automation. Train and fine-tune PyTorch computer vision models, including DINOv2-based vision transformers and R-CNN architectures, and deploy them to AWS SageMaker endpoints for inference.
- Build TypeScript UI components using XState for state orchestration and Apollo Client for GraphQL integration. Implement recursive state machines, callback actors, and deeply nested data workflows. Develop reusable UI systems with Tailwind and styled-components, including filtered image grids, multi-step forms, and hierarchical navigation.

SKILLS

Languages	C#, Python, TypeScript, JavaScript, C++, C, Java, SQL
Backend & Data	.NET, EF Core, Node.js, GraphQL, PostgreSQL, OpenSearch, Kafka, RabbitMQ
Cloud & DevOps	AWS, ECS, EC2, S3, SQS, DynamoDB, CDK, Docker, GitLab CI/CD, GitHub Actions
Frontend & ML	React, XState, Apollo Client, PyTorch, LangGraph, LangChain

PROJECTS

PyTorch Computer Vision App (Java / Python / TypeScript / AWS / LangChain) - GitHub - Live

- Built a full-stack computer vision application using PyTorch Faster R-CNN and Mask R-CNN for object detection and instance segmentation. Developed asynchronous video processing with AWS S3 and DynamoDB for storage and job tracking, supporting progress previews, retries, caching, and synchronized bounding-box and mask overlays.
- Integrated a locally hosted Qwen LLM with LangGraph and LangChain tool calling to translate natural-language requests into validated actions for filtering detections, customizing overlays, and navigating video results.
- Automated Docker builds and deployment to AWS ECS on EC2 Spot using GitHub Actions and AWS CDK. Implemented model switching and persistent LLM prompt caching to manage inference within server memory constraints.

Ski and Rock Climbing E-Commerce Store (.NET / C# / TypeScript / AWS ECS) - GitHub - Live

- Developed a full-stack e-commerce application with product search, filtering, OAuth 2.0, shopping carts, PayPal integration, and order processing. Built a .NET (C#) API using Entity Framework Core and PostgreSQL hosted on AWS RDS.
- Deployed containerized services on AWS ECS Fargate, provisioning infrastructure with AWS CDK, including Application Load Balancers, Route 53, and TLS certificates. Implemented GitHub Actions CI/CD pipelines to automate Docker image builds, ECR pushes, and ECS deployments.

Portfolio Website - Additional Projects (React.js / EC2) - GitHub - Live

- Built a responsive TypeScript portfolio hosted on AWS EC2 with reusable carousels for project screenshots and GIFs.

EDUCATION

B.S. - Computer Science **Mar 2022 - Mar 2024**
Oregon State University - School of Electrical Engineering and Computer Science
Graduated | Major GPA: 4.0 | Honor Roll 2022-2024